

PARAMETERS AND ITS SIZE FOR TRAINING, USE AND MODEL FILE

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	<i>KdotQ</i>	4096	4096	1	16777216	64	MB
MATRIX	<i>Key (K)</i>	4096	64	1	262144	1	
	<i>Query (Q)</i>	4096	64	1	262144	1	
	<i>Horizontal (H)</i>	64	4096	1	262144	1	
	<i>Vertical (V)</i>	64	4096	1	262144	1	
MLP	<i>Hor</i>	64	64	16	65536	0.25	
	<i>Ver</i>	64	64	16	65536	0.25	
Continuous Calculation for every head in full context	<i>KEYS</i>	4096	64	1	262144	1	
	<i>QUERIES</i>	4096	64	1	262144	1	
	<i>EV</i>	4096	64	1	262144	1	
	<i>Hlayers</i>	64	16	2	2048	0.0078125	
	<i>Activations</i>	64	16	2	2048	0.0078125	
	<i>gweights (hor)</i>	64	64	16	65536	0.25	
	<i>gweights (ver)</i>	64	64	16	65536	0.25	
CACHE	<i>QK'</i>	64	64	1	4096	0.015625	
	<i>QV'</i>	64	64	1	4096	0.015625	
	<i>KH'</i>	64	64	1	4096	0.015625	
Complete Model Training Parameters and Size (one parallel is number of attention heads working in parallel)				Attention	18890752	72.0625	
				One Parallel:	151126016	576.5	
				Model:	38688260096	144.125	GB
Very Small	<i>TOKENS</i>	1	64	32768	2097152	8	MB
	<i>1D</i>	3	64	2048	393216	1.5	
TOTAL:				38690750464	144.1342773		GB

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	<i>KdotQ</i>	4096	4096	1	16777216	64	MB
CACHE	<i>QK'</i>	64	64	1	4096	0.015625	
	<i>QV'</i>	64	64	1	4096	0.015625	
	<i>KH'</i>	64	64	1	4096	0.015625	
MLP	<i>Hor</i>	64	64	16	65536	0.25	
	<i>Ver</i>	64	64	16	65536	0.25	
	<i>Hlayers</i>	64	16	2	2048	0.0078125	
	<i>Activations</i>	64	16	2	2048	0.0078125	
For Model Use Parameters and Size				Attention:	16924672	64.5625	
				One Parallel:	135397376	516.5	
				Model:	4332716032	16.140625	GB
Very Small	<i>TOKENS</i>	1	64	32768	2097152	8	MB
	<i>EV</i>	1	64	1048576	67108864	256	
	<i>1D</i>	3	64	256	49152	0.1875	
TOTAL:				4401971200	16.39862061		GB

For Trained Params:	NAME	DIMS			Attention	Model	SIZE (MB)
MATRIX	<i>Key (K)</i>	4096	64	1	262144	536870912	2048
	<i>Query (Q)</i>	4096	64	1	262144	536870912	2048
	<i>Horizontal (H)</i>	64	4096	1	262144	536870912	2048
	<i>Vertical (V)</i>	64	4096	1	262144	536870912	2048
MLP	<i>Hor</i>	64	64	16	65536	134217728	512
	<i>Ver</i>	64	64	16	65536	134217728	512
CACHE	<i>QK'</i>	64	64	1	4096	8388608	32
	<i>QV'</i>	64	64	1	4096	8388608	32
	<i>KH'</i>	64	64	1	4096	8388608	32
MODEL PARAMETERS					1191936	2441084928	9312

Data type	float32
Bytes	4
Embedding	64
Layers (PAs)	8
Blocks (CAs)	8
Attention Per layer (AHs)	32
Layers of MLP	16
Context Window	4096
Height	4096
Number of MLP/Head	2
FULL Context	32768
ATTENTION	Single Head
Partial Attention (Layer)	Consecutive IAs
Block (Complete Attention)	Parallel PAs
Transformer	Multi or Single CA
One Parallel	Heads in parallel ops.
MODEL	Dense

NAME	PARAMETERS	SIZE (MB)	GB
MLP	268435456	1024	1
MAT	2147483648	8192	8
CACHE	25165824	96	0.09375
FOR TRAINING	2415919104	9216	9
FOR USE	293601280	1120	1.09375
TOTAL	2441084928	9312	9.09375